

ArtBench Generative Modeling

Martim Alexandre Vieira Antunes

University of Coimbra, Portugal
uc2025191528@student.uc.pt

Abstract. This work explores generative modeling techniques applied to the ArtBench-10 dataset, with the goal of synthesizing realistic artwork images across multiple artistic styles. We compare three main families of generative models: **Autoencoders** (including Variational Autoencoders), **Generative Adversarial Networks** (DCGAN), and **Diffusion based models**.

During development, experiments were conducted on a reduced subset of the dataset to enable rapid iteration and hyperparameter tuning. The best performing configurations were then trained on the full dataset. Special attention was given to the trade-off between reconstruction quality and latent space regularization in Variational Autoencoders, controlled by the β parameter.

Model performance was evaluated both qualitatively, through visual inspection of generated samples, and quantitatively using **Fréchet Inception Distance (FID)** and **Kernel Inception Distance (KID)**. To ensure statistical robustness, results were averaged across multiple random seeds.

The experimental results highlight the strengths and limitations of each model family in terms of sample quality, diversity, and training stability, providing a comparative analysis of generative approaches in the context of artistic image synthesis.

Keywords: Generative Models · Autoencoders · GANs · Diffusion Models · ArtBench · Image Generation

1 Introduction

Generative models have become a central topic in modern machine learning, enabling the synthesis of realistic data across multiple domains such as images, audio, and text. In computer vision, these models aim to capture the underlying distribution of visual data and generate new samples that are both diverse and visually coherent.

This work focuses on generative modeling of artistic images using the **ArtBench-10 dataset**, which contains paintings from ten distinct artistic styles. Unlike traditional classification tasks, generative modeling requires not only capturing the structure of the data but also producing high-quality and diverse samples that closely resemble the original distribution. [1]

In this project, we explore and compare three main families of generative models: **Autoencoders, including Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs)**, specifically DCGAN, and **Diffusion models**. Each approach presents different trade-offs in terms of training stability, sample quality, and computational cost, providing insights into their suitability for artistic image synthesis.

To ensure a fair comparison, all models are evaluated under a consistent experimental protocol. Performance is assessed qualitatively, through visual inspection of generated samples, and quantitatively using standard metrics such as **Fréchet Inception Distance (FID)** and **Kernel Inception Distance (KID)**. Experiments are repeated using multiple random seeds to guarantee statistical robustness.

The main objective of this work is to analyze the strengths and limitations of each generative approach and identify the most effective model for artistic image synthesis on the ArtBench dataset.

2 Problem Description

The goal of this project is to generate realistic artwork images that reflect multiple artistic styles. The challenge lies in learning the underlying distribution of the ArtBench-10 dataset and producing new samples that are both visually coherent and diverse.

ArtBench-10 contains paintings grouped into 10 distinct styles, with each image having a resolution of 32×32 RGB pixels. Unlike classification tasks, the focus here is on synthesizing data that matches the original distribution in terms of content, style, and variety.

Evaluating generative models requires both qualitative and quantitative assessment: visually inspecting the generated images to ensure they are plausible, and computing metrics such as Fréchet Inception Distance (FID) and Kernel Inception Distance (KID) to quantify similarity to the real dataset.

This project aims to compare different families of generative models, analyze their performance in terms of sample quality, diversity, and training stability, and identify the most effective approach for artistic image synthesis.

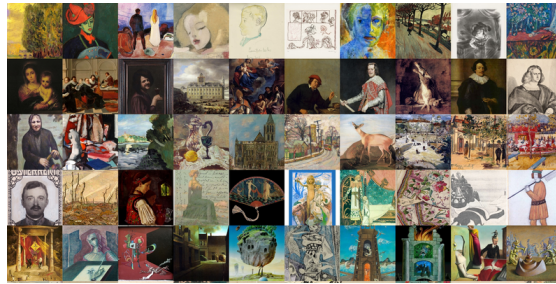


Fig. 1: ArtBench-10

3 Methodology and Approach

This section describes the methodology adopted for implementing and training generative models on the ArtBench-10 dataset. The approach follows a structured pipeline from data loading and preprocessing to model training, sampling, and evaluation.

3.1 Data Representation

For model development, we used the provided 20% subset of the ArtBench-10 dataset. The preprocessing pipeline is designed to ensure consistent image resolution and normalization for all generative models. The main steps are:

- **Resizing and cropping:** Each image is resized to 32×32 pixels using bilinear interpolation and then center-cropped to ensure a consistent square format.
- **Tensor conversion:** Images are converted from PIL format to PyTorch tensors with pixel values in the range $[0, 1]$.
- **Normalization:** Each channel (RGB) is normalized to the range $[-1, 1]$ using mean and standard deviation of $(0.5, 0.5, 0.5)$, suitable for VAE, DCGAN, and diffusion models.

To ensure reproducibility and controlled experiments:

- A subset of the training data was selected using a fixed random seed, allowing us to iterate quickly during model development.
- Data loaders were created with a batch size of 64 and a variable number of workers depending on the runtime environment (e.g., zero workers for notebook kernels to avoid multiprocessing issues).
- The preprocessing and data loader setup ensures that the training, validation, and test splits are consistent across all model families.

Additionally, for the full 50k training images, we used a CSV file specifying exactly which indices to include in the training subset. The remaining images were reserved for evaluation, creating a reproducible train/test split aligned with the experiment protocol.

3.2 Generative Models

We trained three main families of generative models on the ArtBench-10 dataset: Variational Autoencoders (VAEs), Deep Convolutional GANs (DCGANs), and Diffusion models. The training protocol was applied consistently to ensure fair comparison across models.

Variational Autoencoder (VAE) Variational Autoencoders are probabilistic generative models that learn a compressed latent representation of the data while enabling stochastic sampling. The architecture consists of three main components: [2]

- **Encoder:** Convolutional layers with Batch Normalization and ReLU activations progressively reduce spatial resolution, outputting the mean μ and log-variance $\log \sigma^2$ of a Gaussian latent distribution.
- **Latent space:** A continuous multivariate Gaussian space from which new samples can be drawn. The reparameterization trick allows gradient-based optimization through the stochastic sampling step.
- **Decoder:** Transposed convolutional layers map latent vectors back to image space, reconstructing images that match the original data distribution.

Training minimizes a reconstruction loss combined with a KL divergence term weighted by β , which regularizes the latent space towards a standard Gaussian. VAEs produce stable training and smooth latent spaces, though outputs tend to exhibit some blurriness due to the pixel-wise reconstruction objective.

Generative Adversarial Networks (DCGAN) Deep Convolutional GANs [3] are generative models composed of two adversarial networks trained simultaneously in a min-max game. The architecture consists of two main components:

- **Generator:** Takes a random latent vector and progressively upsamples it through transposed convolutional layers with Batch Normalization and ReLU activations, transforming abstract latent codes into structured images.
- **Discriminator:** Downsamples the input image through convolutional layers with LeakyReLU activations, producing a scalar output representing the probability that the input is a real image.

The generator aims to fool the discriminator, while the discriminator learns to distinguish real from generated images. DCGANs are capable of producing sharp and detailed images compared to VAEs, with high visual fidelity. The latent space allows interpolation between samples, although training can be unstable and prone to issues such as mode collapse.

Diffusion Models Diffusion models, specifically Denoising Diffusion Probabilistic Models (DDPMs) [4], are generative models that learn to reverse a gradual noising process applied to images. In the forward process, Gaussian noise is progressively added to an input image over T timesteps. The reverse process predicts and removes the noise step-by-step, reconstructing realistic images from pure noise.

Pixel-space Diffusion: Operates directly on full-resolution images using a UNet-like architecture [5]. The model consists of an encoder (downsampling) with residual blocks, a bottleneck that processes the lowest-resolution features,

and a decoder (upsampling) with skip connections that combine encoder features. Sinusoidal timestep embeddings allow the network to condition its predictions on the current diffusion step.

Generative behavior: Diffusion models are capable of producing highly detailed and diverse images, often surpassing GANs in visual fidelity and stability. The iterative denoising process provides fine-grained control over the image formation, allowing the model to capture complex distributions in the data.

3.3 Evaluation Metrics

To compare the generative models on the ArtBench-10 dataset, we adopt both qualitative and quantitative evaluation strategies, assessing visual quality, diversity, and statistical similarity between generated and real images.

Qualitative Evaluation Generated image grids are inspected visually to assess style fidelity, sharpness, and diversity of outputs across all model families.

Quantitative Metrics Two distribution-level metrics are used for quantitative comparison:

- **Fréchet Inception Distance (FID):** Measures the distance between the distributions of generated and real images in the feature space of a pretrained Inception network. Lower values indicate higher quality and diversity of generated samples [6].
- **Kernel Inception Distance (KID):** Computes the squared Maximum Mean Discrepancy (MMD) between Inception features of generated and real images. Unlike FID, KID is an unbiased estimator and is reported as mean $\pm$ standard deviation over multiple random subsets, providing a more reliable measure for smaller sample sizes [7].

To ensure statistical robustness, the evaluation is repeated across multiple independent runs with different random seeds, and results are reported as mean $\pm$ standard deviation. Full details of the evaluation procedure are described in Section 4.

4 Experimental Setup

This section details the experimental configuration used to train and evaluate the generative models (VAE, DCGAN, and Diffusion Models) on the ArtBench dataset. It describes the protocol, hyperparameter variations, training setup, and evaluation procedures.

4.1 Dataset and Preprocessing

All experiments use the ArtBench-10 dataset, consisting of 50,000 training images across 10 artistic styles at a resolution of 32×32 pixels. During model development and hyperparameter tuning, we used the provided 20% training subset (approximately 10,000 images), selected using a fixed random seed to ensure reproducibility. The best hyperparameter configuration identified for each model family through qualitative analysis on the 20% subset was subsequently used to retrain all three models on the full 50,000-image dataset for final quantitative evaluation.

The preprocessing pipeline is consistent across all model families:

- Images are resized and center-cropped to 32×32 pixels.
- Pixel values are normalized to the range $[-1, 1]$ using mean and standard deviation of $(0.5, 0.5, 0.5)$ per channel.
- No data augmentation was applied to maintain consistency across models.

4.2 Hyperparameter Variations and Model Configurations

Table 1 summarizes the hyperparameter configurations explored for each model family. All models were trained with a batch size of 64 and optimized using the Adam optimizer.

Table 1: Hyperparameter configurations explored per model family.

Model	Latent Dim	Channels	Epochs	Learning Rate	Additional
VAE	32, 128	—	50, 100	1×10^{-3}	$\beta \in \{0.5, 0.7\}$
DCGAN	64, 128	—	50, 100	2×10^{-4}	—
Pixel-space Diffusion	—	128	100	2×10^{-4}	$T \in \{100, 1000\}$, Schedule: linear, cosine

4.3 Training Setup

Hardware and Software:

- **GPU:** NVIDIA RTX 4060
- **CPU:** AMD Ryzen 7
- **RAM:** 8 GB
- **Framework:** Python 3.13.2

Training details common to all models:

- **Batch size:** 64
- **Random seed:** 42, fixed across all experiments for reproducibility
- **Early stopping:** Not applied, to ensure fair comparison across all models at equivalent training budgets

Loss functions:

- **VAE:** Combined reconstruction loss (MSE) and KL divergence, weighted by β :

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{MSE}} + \beta \cdot D_{\text{KL}}(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, 1))$$

- **DCGAN:** Binary Cross-Entropy (BCE) for both generator and discriminator, following the standard minimax objective.
- **Diffusion models:** Mean Squared Error (MSE) between the predicted noise and the true noise.

4.4 Evaluation Protocol

To ensure statistical robustness and fair comparison across all model families, we follow a fixed evaluation protocol. For each trained model:

1. Generate 5,000 images using the trained generator.
2. Sample 5,000 real images from the ArtBench-10 training set.
3. Compute **FID** between the full sets of 5,000 generated and 5,000 real images.
4. Compute **KID** using 50 random subsets of size 100, reporting mean $\pm$ standard deviation.
5. Repeat steps 1–4 for 10 independent runs using different random seeds, keeping model weights fixed.
6. Report final results as mean $\pm$ standard deviation over the 10 repetitions.

The preprocessing pipeline, sample count, and evaluation code are kept identical across all models to ensure a fair and reproducible comparison.

5 Analysis of results

This section presents the experimental results obtained from training and evaluating three generative model families (VAE, DCGAN, and Diffusion) , on the ArtBench-10 dataset. We first analyse each model family individually on the 20% subset to identify the best hyperparameter configuration. The selected configurations are then retrained on the full 50,000-image dataset and evaluated quantitatively using FID and KID.

5.1 Individual Model Analysis (20% Subset)

Variational Autoencoder (VAE) Configuration 1 (latent dim = 32, $\beta = 0.7$, 50 epochs): The KL divergence stabilized around 47.36, with a total loss of 172.87 and reconstruction loss of 139.71. As shown in Figure 2, the generated samples are heavily blurred with poor structural definition, suggesting that a latent dimension of 32 is insufficient to capture the visual complexity of the ArtBench-10 styles.

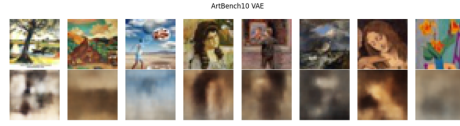


Fig. 2: Samples Configuration 1 - VAE

Configuration 2 (latent dim = 128, $\beta = 0.5$, 100 epochs): The total loss improved to 137.50, with a reconstruction loss of 92.68. The generated samples (Figure 3) show noticeably better structural coherence and color fidelity, though some blurriness remains, which is characteristic of VAE-based models. Configuration 2 was therefore selected as the best VAE configuration.

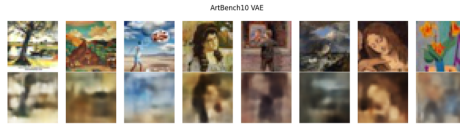


Fig. 3: Samples Configuration 2 - VAE

Generative Adversarial Network (DCGAN) Two configurations of the DCGAN were evaluated on the 20% ArtBench-10 subset.

Configuration 1 (latent dim = 64, 50 epochs): The discriminator loss stabilized quickly around 0.6–0.7, while the generator loss decreased from 5.0 to approximately 3.0 and remained stable throughout training, suggesting a reasonable adversarial equilibrium without evidence of mode collapse. (Figure 4) The generated samples (Figure 5) show some structural definition and color variation across styles, though several images exhibit low contrast and lack of fine detail.

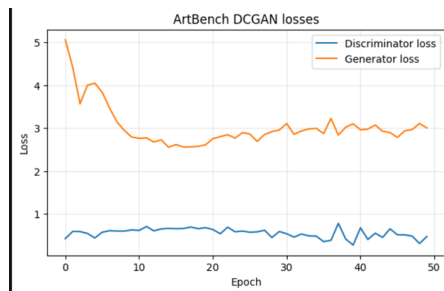


Fig. 4: Loss Configuration 1 - DCGAN

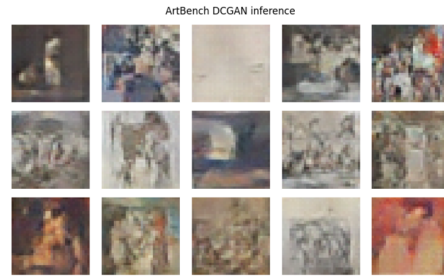


Fig. 5: Samples Configuration 1 - DCGAN

Configuration 2 (latent dim = 128, 100 epochs): The discriminator loss remained low and stable around 0.5, while the generator loss showed a slight upward trend in later epochs (epochs 80–100), reaching approximately 4.0, which may indicate the generator struggling to fool an increasingly confident discriminator. (Figure 6) Despite this, the generated samples (Figure 7) demonstrate notably improved visual quality, with richer color palettes, greater diversity across artistic styles, and more structured compositions. Configuration 2 was therefore selected as the best DCGAN configuration.

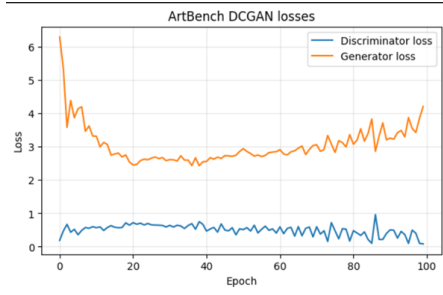


Fig. 6: Loss Configuration 2 - DCGAN

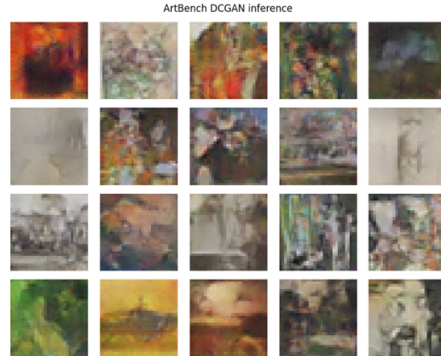


Fig. 7: Samples Configuration 2 - DCGAN

Diffusion Model (Pixel-space) Two configurations of the pixel-space diffusion model were evaluated, both using `model_channels` = 128 and trained for 100 epochs.

Configuration 1 ($T = 100$, linear noise schedule, 100 epochs): The training loss decreased sharply from 0.21 in the first epoch, stabilizing around 0.10 by epoch 100, indicating consistent convergence over approximately 8 hours of training. (Figure 8) The generated samples (Figure 9) demonstrate strong visual quality, with rich and diverse color palettes and recognizable artistic structures across multiple styles.

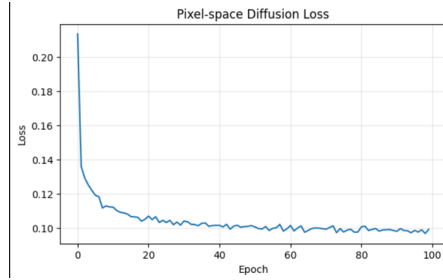


Fig. 8: Diffusion Training Loss Config1



Fig. 9: Samples Configuration 1 - Diffusion

Configuration 2 ($T = 1000$, linear noise schedule, 100 epochs): The training loss decreased from 0.124 in the first epoch, stabilizing around 0.035 by epoch 100, showing consistent convergence and a lower final loss compared to Configuration 1. Although some generated samples (Figure 11) exhibit near-black outputs, the majority show recognizable artistic structures with diverse color distributions, representing an overall improvement over Configuration 1 in terms of loss convergence and sample consistency.

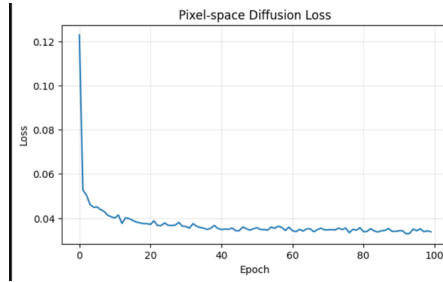


Fig. 10: Diffusion Training Loss Config2

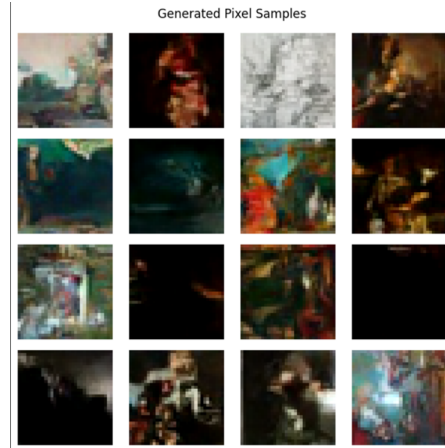


Fig. 11: Samples Configuration 2 - Diffusion

Despite the presence of some collapsed samples, Configuration 2 achieved a lower final training loss (0.035 vs 0.10) and was therefore selected as the best diffusion configuration for final training on the full dataset.

5.2 Full Dataset Evaluation (50k)

All three models were retrained on the full 50,000-image ArtBench-10 dataset using the best configuration identified for each family. Table 2 reports the quantitative results.

Table 2: Quantitative evaluation on the full dataset (mean $\pm$ std over 10 runs).

Model	FID (mean $\pm$ std)	KID (mean $\pm$ std)
VAE	130.400 $\pm$ 0.421	0.118 $\pm$ 0.007
DCGAN	92.385 $\pm$ 0.979	0.0596 $\pm$ 0.009
Diffusion	39.43 $\pm$	0.0253 $\pm$ 0.005

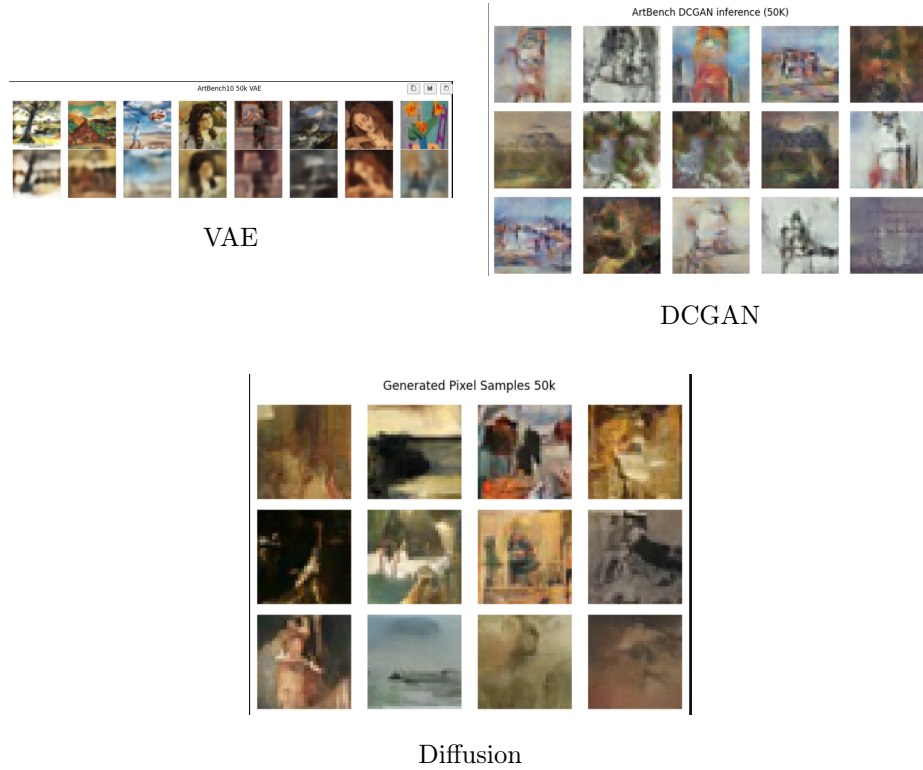


Fig. 12: Generated samples from VAE, DCGAN, and Diffusion trained on the full dataset.

5.3 Discussion

The quantitative results in Table 2 demonstrate a clear ranking among the three generative model families. The VAE achieved the weakest results ($\text{FID} = 130.40 \pm 0.42$, $\text{KID} = 0.118 \pm 0.007$), consistent with its tendency to produce blurry outputs, though the low standard deviation confirms stable and reproducible generation.

The DCGAN improved significantly ($\text{FID} = 92.39 \pm 0.98$, $\text{KID} = 0.060 \pm 0.009$), and notably outperformed the diffusion model during the 20% subset experiments, suggesting that adversarial training is particularly effective under limited data regimes.

When retrained on the full 50,000-image dataset, the diffusion model surpassed all others ($\text{FID} = 39.43$, $\text{KID} = 0.025 \pm 0.005$), demonstrating that diffusion models benefit more substantially from increased data availability than GANs. Due to computational constraints, the diffusion evaluation was conducted with only 3 seeds instead of 10, so the FID standard deviation could not be reliably computed and results should be interpreted accordingly.

Overall, the results confirm the expected trade-offs: VAEs offer stable training at the cost of visual fidelity; DCGANs perform competitively under limited data; and diffusion models achieve the highest quality given sufficient data, at the expense of significantly higher computational cost.

6 Conclusion

This work presented a systematic comparison of three generative model families, VAEs, DCGANs, and Diffusion models, on the ArtBench-10 dataset. Results confirmed the expected trade-offs between stability, visual fidelity, and computational cost. Notably, while the DCGAN outperformed the diffusion model on the 20% subset, the diffusion model achieved the best overall results on the full dataset ($\text{FID} = 39.43$, $\text{KID} = 0.025 \pm 0.005$), highlighting the importance of data availability when selecting a generative approach.

The VAE, despite its training stability, consistently produced the weakest results due to its blurry outputs, while the DCGAN demonstrated competitive performance particularly under limited data conditions. The diffusion model, although computationally demanding, proved to be the most capable of capturing the complexity and diversity of the ArtBench-10 artistic styles when given sufficient training data.

Limitations of this work include the computational constraints imposed by the available hardware, which restricted the diffusion model evaluation to 3 random seeds instead of the required 10, and limited the exploration of larger architectures and longer training schedules. Future work could explore more efficient sampling strategies to reduce inference time, as well as conditional generation to produce style-specific artwork.

References

1. Liao, P., Li, X., Liu, X., Keutzer, K.: The ArtBench Dataset: Benchmarking Generative Models with Artworks. arXiv preprint arXiv:2206.11404 (2022)
2. Weng, L.: From Autoencoder to Beta-VAE. Lil'Log (2018). <https://lilianweng.github.io/posts/2018-08-12-vae/>
3. Radford, A., Metz, L., Chintala, S.: Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks. arXiv preprint arXiv:1511.06434 (2015)
4. Ho, J., Jain, A., Abbeel, P.: Denoising Diffusion Probabilistic Models. Advances in Neural Information Processing Systems (NeurIPS) (2020)
5. Ronneberger, O., Fischer, P., Brox, T.: U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI (2015)
6. Fréchet Inception Distance, Wikipedia.
7. Distributional Metrics: KID. apxml.com (2024). <https://apxml.com/courses/synthetic-data-gans-diffusion/chapter-5-evaluating-synthetic-data-quality/distributional-metrics-kid>